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Recent Advancements, Challenges, and Future Directions of Integrating Drones in Smart City Traffic Management: A Systematic Review

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ABSTRACT

The implementation of unmanned aerial vehicles (UAVs) or drones has gained immense attention in recent years due to their high potential in enhancing traffic management in a smart city context. This systematic review aims to provide a thorough analysis of the applications of drones in smart traffic management, and limitations and challenges upon technology adoption, as well as future research directions in this field. The search was conducted by following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) method to identify relevant articles based on inclusion and exclusion criteria published in the past 10 years, specifically from January 1, 2014, to September 19, 2024. By analyzing the final selection of 14 research articles from databases such as Scopus, ScienceDirect, SpringerOpen, ACM Digital Library, and EBSCO, the results reveal that integrating UAVs into smart city traffic management provides substantial advantages in the perspectives of both ground monitoring and airspace control. This systematic review also contributes to offering a solid foundation for researchers or stakeholders to gain a thorough understanding of drone technologies in making revolutionary breakthroughs within the field of traffic-related aspect to strengthen the transportation system in smart cities.

1 | Introduction

1.1 | Smart Cities and Traffic Management

The future of urban development seems to lie in the concept of a "smart city." Although the definition is not set in stone, a popular interpretation is that a smart economy, smart mobility, smart environment, smart people, smart living, and smart governance are the six necessary features for a smart city [1]. These features are the primary factors when it comes to measuring the competitiveness of urban spaces, and by utilizing new technologies, these factors can be upgraded into their "smarter" forms. According to Ref. [2], the goal of smart cities and their respective improved features is urban sustainability. This is in line with the UN Sustainable Development Goals (SDG) as Goal Number 11: Sustainable Cities and Communities. The process of integrating sustainability into smart mobility, an aspect of smart cities as previously stated, is through improvements in five key features: accessibility, safety, environment-friendliness, efficiency, and quality of life. Traffic is an especially vicious problem in most

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modern cities, irrespective of geographical location and economic development level. Therefore, traffic management is a necessity moving forward in the path to sustainable cities. Utilizing the benefits of something like Internet of Things (IoT) can allow for smarter management through area wide and real-time knowledge of the state of traffic [3].

1.2 | Drones in Smart Traffic Management Cities

A potential new technology being explored to meet city sustainability goals is unmanned aerial vehicles (UAVs) or better known as drones. Drone technology has already seen extensive use in military applications [4] and is also becoming more commonplace as tools for hobbyists [5]. At this moment in time, drone technology is widely considered for large-scale use in one of two civil roles within urban areas: first, remote sensing devices and second, delivery vehicles.

When it comes to acting as remote sensing devices, drones are popular to be applied in various sectors including military, photography, land surveillance, agriculture, and recreational use [6–10]. Urban environments are a perfect place for the deployment of drones, as their many unique qualities allow them to be used for seeking points of traffic congestion in large areas with relative ease.

Delivery drones have not been developed as much as sensing drones yet, but there are several large corporations that are running pilot programs to see the viability of drones as a medium of delivery, such as Google, Amazon, Alibaba, and many more companies. With increasing demand for delivery services and ever-growing urbanization, the need for improved localized logistics systems is pushing the creation of delivery drone networks.

In the field of traffic management, drones are typically envisioned into various small divisions. Their aerial perspective and deployment make them suitable for tasks such as vehicle detection, counting and tracking related tasks [11, 12] as well as road accident monitoring, detection, reconstruction, and reporting [13]. With the application of drones in road accident management, accidents can be handled with great efficiency without causing disruption or hindering the normal flow of traffic while at the same time reducing the need for a person in charge and their injury risks and enabling the collection of real-time information easily. Systems like MultEYE [14] illustrate how UAV-based traffic monitoring can defeat traditional roadside cameras by ensuring real-time observation under real-world constraints. As a whole, these use cases emphasize drones as an important enabling technology for smart cities for addressing the growing challenges of traffic safety, congestion, and efficient urban mobility.

1.3 | Significance of Study

The integration of drones in smart city traffic management provides a paradigm shift in conventional traffic management. It has significant potential to address many urban mobility challenges due to the growing amount of traffic in current cities. Although drones are applied in many different fields, there is still a lack of comprehensive systematic review in assessing the integration of drones in smart city traffic management. Hence, this paper is written with the aim of synthesizing the existing knowledge on the applications, challenges, and the future directions or improvements of smart cities' traffic management through the analysis of findings and insights from various studies and providing more valuable insights into urban planners or the government in developing a smart traffic management city.

1.4 | Aim and Research Questions

The aim of this systematic review paper is to synthesize the existing knowledge on the applications, challenges, and future directions or improvements of smart cities' traffic management by analyzing findings and insights from various studies. This systematic review also aims to provide a comprehensive understanding of how drones could be employed to enhance urban planning and investigate any research opportunities for further improvement and development. Hence, the research questions of this systematic review are outlined as follows:

- RQ1: What are the primary applications of drones integrated in smart cities, and how do they contribute to urban development?
- RQ2: What are the challenges and limitations of implementing these drone technologies?
- RQ3: What future improvements can be made to improve urban development?

1.5 | Structure of Paper

The systematic review is further divided into five sections, with Section 2 outlining the overview and requirement of research methodology collecting relevant publications to be included in this review. Section 3 discusses the results focusing on both qualitative and quantitative analysis methods to prove the suitability and level of bias of the selected studies. Section 4 outlines the discussion on applications of drones in smart cities, along with the challenges and future improvements to better enhance urban city planning. Finally, Section 5 concludes the systematic review with a summary of findings and contributions of the review.

2 | Research Methodology

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were strictly adhered to in this systematic review to make sure all the phases of the research process are carried out in a transparent way. The PRISMA 2020 statement has been a preferred reporting framework in writing a systematic review as it is able to improve the clarity of reports with its structured approach, which could further enhance the reliability of the results in report writing [15]. Figure 1 shows the PRISMA flow diagram that shows each phase of the study selection process to give a glance at the number of articles included in this study.

2.1 | Database Selection

To capture the search results as far as possible, the first phase of this systematic review started off with identifying potential databases that could return publication results that cover the drone applications in smart cities specifically in terms of transportation aspect. A total of 12 candidate databases were listed due to their vast number of technical-related publications: IEEE Xplore,

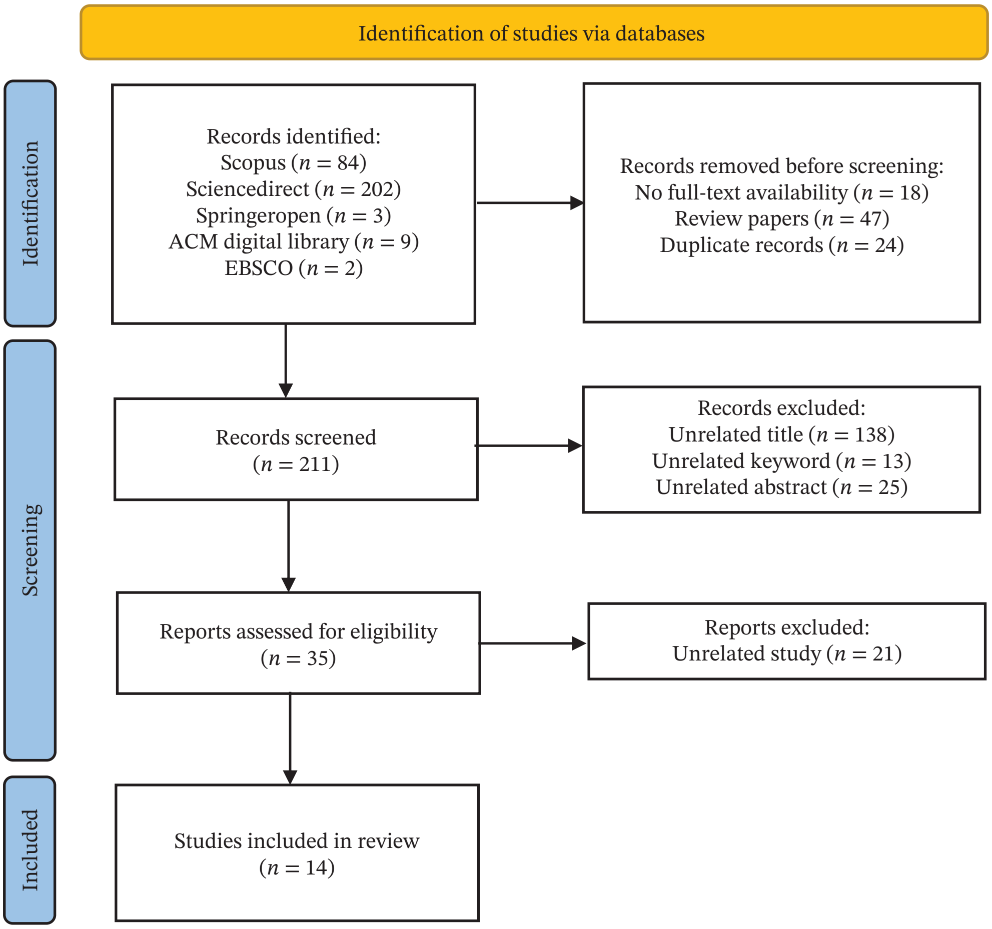

FIGURE 1 | PRISMA flow diagram.

Scopus, Science Direct, Google Scholar, Web of Science (WoS), Tun Hussein Onn Sunway Library, SpringerOpen, Association for Computing Machinery (ACM) Digital Library, ProQuest, Elton B. Stephens Company (EBSCO), Directory of Open Access Journals (DOAJ), and Multidisciplinary Digital Publishing Institute (MDPI).

Nevertheless, to ensure an efficient and rigorous search process, the selection criteria emphasized the databases with high indexing coverage and concise search capabilities. A systematic review ideally seeks to maximize coverage while minimizing the redundancy. Hence, Scopus was selected as the primary database as it serves as an aggregator indexing the majority of technical publications found in IEEE Xplore, WoS, and Springer. As a result, by utilizing Scopus alongside ScienceDirect, ACM Digital Library, SpringerOpen, and EBSCO, this systematic review captures high-impact studies from the broader engineering landscape to ensure relevance. This also allows for better literature retrieval without searching overlapping indexes to avoid redundancy.

2.2 | Search Strategy

In terms of the search strategy in this review article, the advanced search strings were designed using the combination of keywords from multiple themes such as drone, smart city, transportation, and logistics, which are connected with Boolean operators AND and OR. Wildcards symbols specifically asterisk (∗) were also used in constructing advanced search strings to capture the area of results returned as much as possible by matching the relevant characters. The initial selection of the advanced search strings to be used in databases is shown in Table 1.

A systematic search strategy was developed by deriving the main themes and their related synonym as presented in Table 1. There were three primary themes of focus: (i) drone technology and general traffic management, (ii) drone-based accident detection and vehicle tracking, and (iii) finally, drone applications in logistics and traffic congestion management. Due to limitations of certain databases in terms of search string length and usage of Boolean operators, the search strings were then refined to only select the most relevant keywords from the initial keyword list. As a result, the following three advanced search strings were carefully constructed and used to collect publication results within selected databases, with all searches conducted on 19 September 2024.

1. (“Drone∗” OR “Unmanned Aerial Vehicle∗” OR “Remotely Piloted Vehicle∗”) AND (“Traffic Management” OR “Transport Management” OR “Road Condition Monitoring”) AND (“Smart Cit∗”)
2. (“Drone∗” OR “Unmanned Aerial Vehicle∗” OR “Remotely Piloted Vehicle∗”) AND (“Accident Detection” OR “Vehicle Tracking”) AND (“Smart Cit∗”)
3. (“Drone∗” OR “Unmanned Aerial Vehicle∗” OR “Remotely Piloted Vehicle∗”) AND (“Logistics Management” OR “Traffic Jam” OR “Congestion”) AND (“Smart Cit∗”)

TABLE 1 | Initial Selection of Advanced Search String Formation.

Themes	Synonyms/similar terms
Drone	• Drone • UAV • Unmanned aerial vehicle • Camera-mounted drone • RPV • Remotely piloted vehicle • RPAS • Remotely piloted aircraft system
Traffic management	• Ground traffic management • Transport management • Traffic monitoring • Traffic handling
Road accident detection	• Traffic accident detection • Car accident detection • Road accident
Vehicle counting and detection	• Vehicle tracking • Motor vehicle tracking • Automobile tracking
Logistics	• Logistics management
Road intersection monitoring	• Road condition monitoring • Traffic monitoring
Traffic congestion	• Traffic jam • Congestion
City	• Smart city
AI/Techniques	• Artificial intelligence • Machine learning • Computer vision • Visual-based navigation system • Deep learning • Autonomous

Table 2 shows the number of publication results returned within the five selected databases. Based on the search results, a total of 300 publications were taken into consideration for the following process.

2.3 | Inclusion and Exclusion Criteria

The following inclusion and exclusion criteria were formed to ensure consistency and eligibility for results:

1. Inclusion Criteria
   - Open access publications with full-text PDF accessibility
   - Studies published between 2014 up to the date of search (September 19, 2024)
   - Papers written in the English language
   - Publication types of journal articles, research articles, and conference papers
2. Exclusion Criteria
   - Publications without full-text PDF accessibility
   - Studies published before 2014
   - Non-English publications
   - Review papers
   - Duplicated records
   - Conceptual or policy-based studies

Upon the development of both inclusion and exclusion criteria, 89 records were excluded from the initial selection process as shown in the depicted PRISMA flow diagram in Figure 1, with 18 of the results having no full-text PDF accessibility, 47 of them consisted of review papers, and 24 duplicated records across the databases. A total of 211 selections were carried forward to the next step for further screening.

2.4 | Screening Process

The screening process carried out in this systematic review was divided into four primary steps. It aims to remove unrelated content aside from the applications of drones in smart cities specifically in the transportation aspect. The screening process was performed from October 15, 2024, to October 22, 2024. Table 3 shows the division of the screening process conducted in this review article.

The screening process was applied to the 211 selections included in the previous process. Out of the selections, 176 results were excluded due to unrelated study about drones, smart city, and transportation aspect, with 138 of the results being excluded from title screening, 13 being excluded due to unrelated keyword, and 25 being removed from abstract screening. Finally, in the full-text review, papers were assessed for technical validity. To ensure the review focused on actionable advancements, studies that lacked empirical validation, such as simulation results, prototype testing, and field experiments were excluded. This filtering ensured that the final selection of 14 articles represented the proven technical implementations rather than detailing concepts. Hence, 21 out of 35 papers were excluded as the methodology and result did not meet the primary aim of this study. As a result, 14 articles were chosen from the screening process for further analysis in the following chapter. This corpus was acquired through a transparent and multistage filtering process, which represents the core themes and advancements in the integration of drones for smart city traffic management.

2.5 | Quality Assessment and Risk of Bias

To ensure the reliability of this systematic review, a quality assessment was implemented into the selection process according to PRISMA reporting standards. A criterion-based quality assessment was employed. During the full-text review, studies were evaluated based on three quality criteria:

1. Clarity of methodology, where the algorithm, model, or simulation of the study must be clearly defined
2. Validity of results, where quantifiable performance metrics must be presented
3. Relevance to traffic management, where the application must directly address traffic flow or logistics within a smart city context

TABLE 2 | Detailed breakdown of results returned.

Databases	String 1	String 2	String 3	Total
Scopus	38	24	22	84
ScienceDirect	0*	21	181	202
SpringerOpen	2	0	1	3
ACM Digital Library	4	0	5	9
EBSCO	50,189*	1	1	2
*Excluded	TOTAL	300		

*The results excluded in the study.

The article screening process was done on a team basis with three reviewers involved in filtering the selection. Any disagreements that arise during the process were resolved by having constant discussion among reviewers to further minimize bias while increasing the reliability of the final results.

3 | Results

3.1 | Study Selection

The final selection of 14 articles represents the core of currently available high-quality and peer-reviewed literature specifically integrating UAVs, smart city infrastructure, and traffic management. While the initial search returned a high volume of results, the rigorous screening process excluded the conceptual papers, policy reviews, and studies lacking technical validations. The resulting sample size reflects the nascent nature of this domain. Drone-related research is abundant; its specific application for integrated smart city traffic management, however, is an emerging field. This study selection ensures the review analyzes only the valid technical implementations and viable frameworks rather than theoretical concepts.

Table 4 shows the final selection of 14 papers with their respective details. Out of the 14 selections, a detailed breakdown of publications years based on the final selection of articles is shown in Figure 2. Referring to the chart, the interest in the application of drones in traffic management started to grow from the year 2020 with a significant increment of publications from that year onwards. The number of articles published in the years 2020 and 2023 was the highest, and this may be due to the increasing technological advancement in automations, artificial intelligence (AI), and IoT technologies [30]. This indicates that this field has gained immense attention with more research conducted related to this area. In contrast, no publications were selected in the range of 2016–2018 in this review article, which may indicate limited research exists in this field within the mentioned period.

3.2 | Knowledge Mapping and Visualization

The 14 selected papers formed the basis of constructing the systematic review to further explore the advancement, challenges, and future improvement of drone technologies in smart city traffic management. A bibliometric network diagram was developed via VOSviewer as shown in Figure 3 to represent the interconnections of selected studies. The bibliometric network diagram serves as a useful visualization to identify the major research themes and trends within the research landscape.

In the figure, there exists several major research themes or clusters that cover the primary domains expressed in the scope of this study, such as the drones' applications in intelligent transportation system with respective technologies used and their integration into smart cities' context. The light green-colored clusters mainly depict the usage of drones including vehicle detection, urban transportation, and smart traffic management, as well as the technologies involved, for instance, AI, remote sensing, and deep learning.

Apart from that, the color scheme presented in the bibliometric diagram shows the frequency where the topics being mostly discussed during a certain period. The more a keyword transforms from blue to yellow, the more recent research focus is within that area. As presented in the figure, the research topics of "smart city" and "vehicle detection" are getting more attention within the research field recently. This may be due to the emerging trend of different object detection algorithms introduced from time to time specifically in the context of deep learning [31, 32] such as convolutional neural network (CNN), you only look once (YOLO), and single shot detector (SSD).

TABLE 3 | Breakdown of screening process.

Screening process	Description
Title screening	Articles selected from previous process were initially screened based on titles to remove irrelevant studies. Only selected articles are proceeded to the second screen
Keyword screening	Results that passed the title screening were evaluated based on keywords. Selected articles should contain keywords such as (i) Keywords related to drones (ii) Keywords related to smart cities (iii) Keywords related to any field of traffic aspect
Abstract screening	Articles passed the keyword screening were screened based on the abstract to make sure all the selected articles have consistent content. Selected articles were carried forward to final screen
Full-text review	Results that passed the abstract screening were finally subjected to a full-text review to examine and evaluate the study in detail in terms of methodology and result sections. Selected articles were included in the final selection

TABLE 4 | Articles included in final selection.

No	Article	Authors	Year
1	[16]	Tahir et al.	2024
2	[17]	Nguyen et al.	2021
3	[18]	Roberts et al.	2020
4	[19]	Moshayedi et al.	2023
5	[20]	Garau Guzman et al.	2024
6	[21]	Borghetti et al.	2022
7	[22]	D'Aloia et al.	2015
8	[23]	El-Sayed et al.	2019
9	[24]	Iranmanesh et al.	2020
10	[25]	Doole et al.	2021
11	[26]	Shawky et al.	2023
12	[27]	Ragab et al.	2022
13	[28]	Qin and Pournaras	2023
14	[29]	Juan et al.	2020

Terms such as "artificial intelligence" and "vehicle detection" also serve as a connector to link traditional UAV applications and recent topics like smart cities. This reflects that these technologies are being integrated progressively into urban city planning nowadays. The network diagram depicts the relevance and intention of the final selection to be included in this review article by discovering the research topics on trend, while it also reveals the authors' rationale of constructing this systematic review with minimal bias in the results.

3.3 | Qualitative Data Analysis

A comprehensive qualitative data analysis was also carried out to ensure the relevance of all 14 selected studies. A column chart showing the most frequently appeared terms was created as presented in Figure 4 by utilizing the text mining functionality integrated in the R data analysis language. The word cloud was generated by displaying the top 150 most frequently occurring terms. The text mining functionality integrated in R data analysis language serves as a useful tool to provide readers with a glance at areas of interest and primary research themes involved in the studies. The height of the terms displayed represents its respective frequency across all included articles. The larger the term displayed in the diagram, the higher its frequency appears across the studies. As depicted in the diagram, keywords such as "drones" and "traffic" take up the dominant portion of the column chart, which may indicate that these terms appeared in most or all studies. This visualization provides a credible indication that all the selected studies consist of similar levels of bias, thereby ensuring the quality of the reviewing and filtering process carried out by the reviewers.

3.4 | Categorization of Drones in Smart Cities Traffic Management

The applications of drones in smart cities can be further divided into four subcategories after carefully reviewing the various application areas that appeared in the final selection of studies. Along with the corresponding references as shown in Table 5, the four subcategories are listed as follows:

- UAV-based object detection
- Flight trajectory and drone traffic control
- Intelligent traffic management and monitoring
- Drone logistics and delivery

3.5 | Key Findings of Selected Studies

The studies selected in this review article describe various applications of drones in smart city traffic management. In extension to the categorization as presented in the previous section, this section outlines the major findings for each selected article from the categories which depicts the methodologies, challenges, and future works across the examined studies as detailed in Table 6, while Table 7 quantifies a systematic performance comparison across the selected studies in terms of accuracy/error, efficiency/speed, scalability/capacity, resource consumption, and finally dataset and scenario details.

A critical analysis of the methodologies (Table 6) and performance metrics (Table 7) show various trends and trade-off within current research landscape. In terms of accuracy–efficiency trade-offs in object detection, the performance data highlights a clear divergence in optimization priorities, such as maximizing detection accuracy by employing heavy architectures [16] and utilizing lightweight models to achieve high frame rates [19]. This suggests current hardware constraints force researchers to choose between precision for surveillance and latency for autonomous navigation.

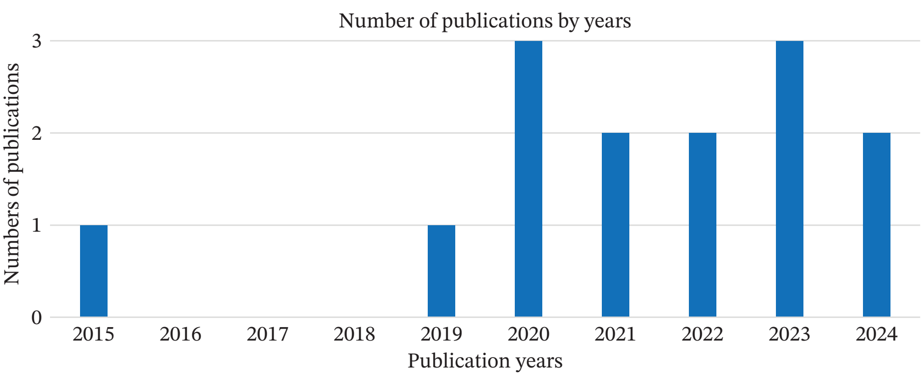

FIGURE 2 | Number of publications on drone applications in traffic management by year.

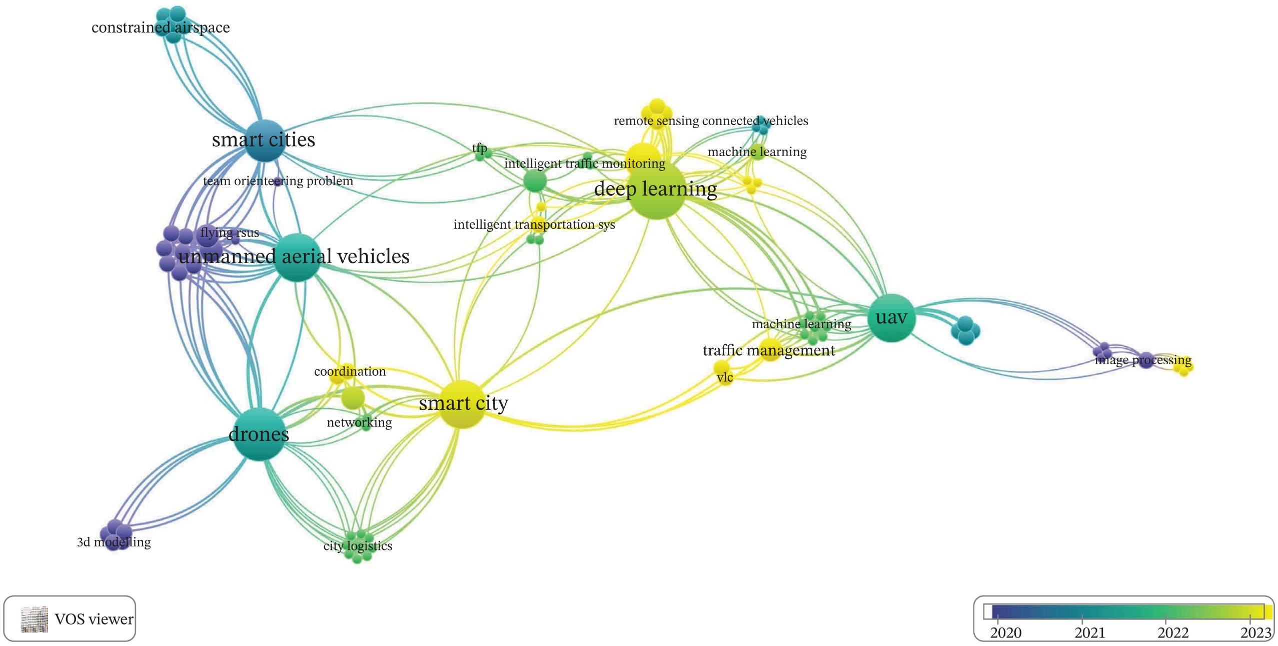

FIGURE 3 | Bibliometric network diagram via VOSviewer.

Furthermore, a cross-comparison of resource consumption metrics reveals a significant reporting gap. Studies related to the drone swarm category [27–29] consistently quantify energy consumption as a primary performance metric, recognizing battery life as the critical constraint for network continuity. However, studies in the object detection category [16, 19, 34, 35] rarely report energy consumption, focusing almost exclusively on mAP or FPS. This shows a disconnection in the literature that the impact on drone's limited flight time remains largely unquantified despite its increasing algorithm accuracy.

Table 6 shows a methodological shift from traditional heuristics to deep learning and swarm intelligence. Early approaches in parking and pavement detection [18, 36–38] relied on static image processing offering high reliability but require specific conditions. Recent advancements [24, 39] move toward adaptive deep learning and dynamic routing algorithms to handle the unpredictability of smart city environments.

4 | Discussion

According to the final selections, the applications of UAVs are categorized into two main categories, namely, road (land) traffic management as well as drone (airspace) traffic management as presented in the complete visualization in Figure 5. Each category is further divided into several subcategories, which

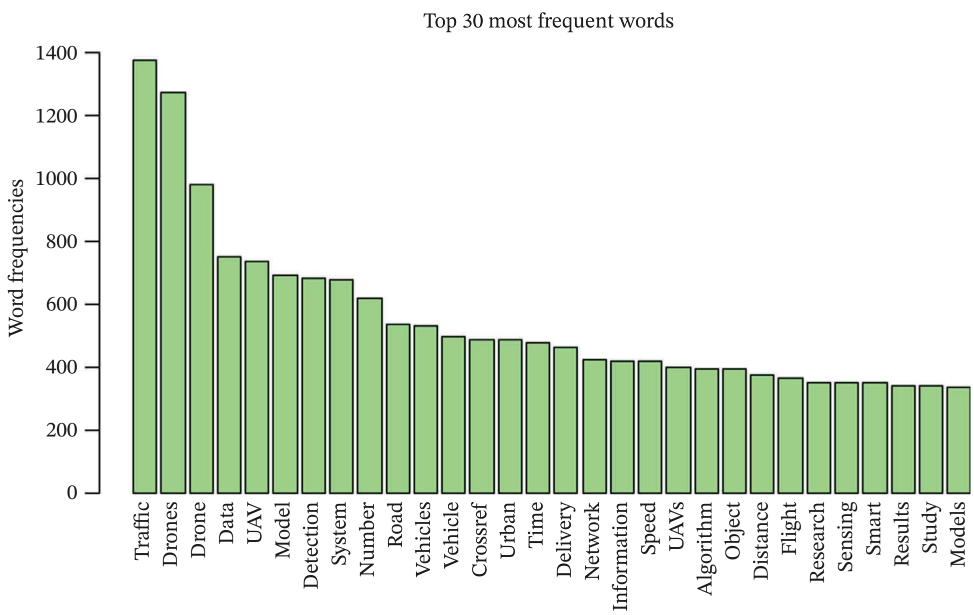

FIGURE 4 | Most frequently appeared terms across selected studies via R Studio.

TABLE 5 | Subcategories of drone applications in smart cities.

Application areas	References
UAV-based object detection	[16] [19] [22] [26]
Flight trajectory and drone traffic control	[17] [27] [28] [29]
Drone logistics and delivery	[21] [24] [25]
Intelligent traffic management and monitoring	[18] [20] [23]

demonstrate the capability and functionalities of UAVs in smart cities. It is noted that while computer vision techniques (e.g., YOLO, CNN) appear across multiple categories, they are analyzed differently as a core detection capability in Section A and as a component of integrated management systems in Section C. This chapter focuses on discussing various applications of integrating drones in smart city traffic management, the challenges while implementing these technologies, and the authors' suggestions toward future improvement.

4.1 | UAV-Based Object Detection

The employment of UAVs in the field of object detection has proven to be a revolutionary milestone in a way that opens up a diverse range of applications in smart city traffic management. In this review, this category specifically focuses on algorithmic capabilities and performance metrics of drones acting as visual sensors. The studies analyzed in this section primarily aim to enhance the precision, speed, and reliability of identifying specific entities, such as pedestrians, vehicles, and accidents. This improves the drones' fundamental ability to "see" the environment. UAVs serve as videography and photography devices, which gather necessary information that can then be further processed using advanced image detection algorithms. This subchapter delves into how surveillance drones are capable of resolving traffic-related challenges along with their detailed methodology and evaluations.

As compared to other alternatives, drones have become a popular tool in capturing high-quality footage, especially in the aspect of traffic management. Drone footage is valuable with its ability to not only capture information such as traffic density and vehicle average speed but also detect concise vehicle trajectory that involves the lane changing and examining drivers' behaviors [40]. On the other hand, there are numerous object detection algorithms including YOLO, SSD, and R-CNN being introduced from time to time. These algorithms form the foundation in detecting and classifying objects under various scenarios by employing them in different domains such as agriculture, disaster management, military, medical, and autonomous driving [5, 32, 39]. These algorithms enable researchers and practitioners to further develop and fine tune the models to detect roadway objects more accurately, for instance, occlusion, diverse illumination, and varying traffic densities across different time periods. Hence, they are becoming important initiatives for enhancing the effectiveness of UAV-based traffic management systems nowadays.

4.1.1 | Pedestrian and Vehicle Detection

The detection of pedestrians and vehicles represents a foundational capability of UAVs in smart cities, yet researchers have adopted various methodologies to address the challenges of aerial monitoring. Tahir et al. [16] and Khemmar et al. [34] employ different deep learning frameworks but prioritize conflicting operational constraints. Tahir et al. focused on maximizing the detection accuracy in dense traffic conditions, especially addressing the occlusion issue where small objects are blocked from view. In this study, the author proposed a pedestrian and vehicle detection network (PVSwin-YOLOv8s) by integrating a Swin Transformer block and Soft Non-Maximum Suppression (Soft-NMS) into the YOLOv8 architecture, and they successfully refined feature extraction to detect occluded vehicles that traditional CNN models often missed. While this approach achieved a 4.8% improvement in mAP, it contrasts the approach taken by Khemmar et al. focusing on computational speed for a real-time embedded system by benchmarking the deep learning models such as YOLO and SSD against traditional methods with improved histogram oriented gradient (HOG) and deformable part model (DPM) detection. The performance reveals that faster-DPM has shown a decline in computational time as compared to the classic DPM, with YOLO and SSD achieving good results in terms of qualitative evaluation.

However, a critical limitation shared by both Tahir et al. and Khemmar et al. is their reliance on the visual spectrum, which restricts the operations to daylight hours. Ma et al. [41] diverge from this standard approach by employing low-resolution thermal imagery using UAVs to address the gap in night-time monitoring. Unlike the bounding box regression used in the visual studies, Ma et al. employed blob extraction and support vector machine (SVM) classifiers to distinguish the pedestrians based on heat signatures. This highlights a complementary relationship in the literature that while deep learning visual models excel in complex daytime traffic, the thermal-based approaches are still necessary for a long-duration surveillance capabilities where the visual systems fail due to low light. These studies have discovered different methods in UAV-based detection and tracking systems while also attempting to bridge the gap identified in other studies. However, there is a common difficulty mentioned in the studies in detecting small and occluded objects [16, 39, 41], which may indicate a gap for future research to be carried out in this area.

4.1.2 | UAV-Based Speed and Accident Detection

UAVs play an important role in vehicle speed enforcement for their smart detection and tracking abilities. This is specifically shown in the domain of speed enforcement and accident management in the reviewed studies. Shahbazi et al. [35] and Moshayedi et al. [19] employ fundamentally different mechanisms for vehicle tracking. Shahbazi et al. rely on the photogrammetric 3D constructions and CNN that prioritize high precision in complex road geometrics like roundabouts and highways, where the models achieved an F1 score of 94.54%. Conversely, Moshayedi et al. prioritize the processing speed over

TABLE 6 | Key findings from selected studies.

References	Aim	Gap/problem	Methodology	Findings	Future work
UAV-based object detection					
[16]	To enable mobile and cost-effective UAVs for efficient traffic management through reliable pedestrian and vehicle detection	CNN-based detectors often struggle with occlusion in UAV object detection	YOLOv8 with transformer for global features, CBAM for missed detections, and Soft-NMS for double detection. Performance compared against YOLOv8 model, older YOLO models, and non-YOLO models	The results showed that the performance outperforms all its competitors. All added enhancements contributed to improving the model's performance	Improvement on detecting small vehicles, improving detection accuracy while reducing resource consumption
[19]	To propose a more reliable and secure UAV-based vehicle speed detector	There were many reported cases of accident damages and injuries for police officers as well as fixed traffic cameras	Detector MobileNet (SSD-MobileNet) was used as the deep learning model to detect vehicles, improved VASCAR for speed estimation, and web-based GUI. Tested multiple UAV altitudes and camera angles	Higher altitudes improved accuracy; specific angles ($-15°$) reduced detection errors	Evaluate additional camera angles, compare results with fixed cameras and explore swarm-based speed averaging
[22]	Propose a method to detect parking slot occupancy using UAV camera images	Increasing vehicle numbers lead to parking issues, and current hardware-based detection methods face high costs and obstacles in the field of view	Analyze reference markers, segment images, perform edge detection, contour shaping and morphological processing, and match profiles to determine slot availability	The system achieved a cross-correlation coefficient of more than 0.9916	Expand the methodology by developing a real-time parking slot monitoring system
[26]	Investigate traffic safety at roundabouts using drones and video processing to identify vehicle conflict points	Traditional safety studies rely on crash reports, which are less reliable in LMICs. There is a need to understand driver behavior and geometric safety at roundabouts, especially where rules are not consistently followed	Used drones to collect 50 min of footage in Cairo and applied object detection and MOT with DCF-CSR algorithm, and improve conflict point using postencroachment time (PET)	Intersection geometry and driver behavior impact safety. Aggressive behaviors like tailgating and rule violations are key contributors	Implement proactive measures like Road Safety Audits Develop real-time conflict monitoring systems Strengthen driver education

(Continues)

TABLE 6 | (Continued)

References	Aim	Gap/problem	Methodology	Findings	Future work
Flight trajectory and drone traffic control					
[17]	To propose an urban airspace design and autonomous drone management	Lack of structured air traffic rules for UAVs, challenges with wind and air turbulence, and collision avoidance	Fixed trajectories using inversion-model-based control, Makarov model for drone following, safe distance model improvements, UAV dynamics for landing deceleration, and CbDMS for monitoring	Proposed method shows efficiency in adverse wind conditions and GNSS limitations. CbDMS improved by GPS updates or Kalman filters	More situational testing needed Makarov model evaluation, enhance drone-following model for broader interactions, and apply to complex urban landings
[27]	Design a Swarm Intelligence Route Selection Strategy (SIRSS-CIoD) for smart city drones	Limited computational power and battery life affect data transmission efficiency; mobility and dynamic topology pose communication challenges	Tuna swarm algorithm for energy-efficient clustering and biogeography-based optimization for route selection to improve data transmission	Improved network lifetime, delay, energy use, throughput, and packet loss compared to existing methods	Develop data aggregation and scheduling methods for more efficient Internet of Drones (IoD) operations
[28]	Propose a decentralized, energy-efficient model for drone swarms in spatiotemporal sensing tasks	Centralized methods suffer scalability and energy efficiency issues, ignoring constraints of decentralized task assignments and battery life limitations	Economic planning and optimization selection (EPOS) algorithm for task assignment, enabling decentralized decision-making based on shared information	Increased energy efficiency and precision (46.45% higher accuracy, 2.88% better efficiency in detecting vehicles)	Extend to smart city applications (disaster response, smart farming); explore collision avoidance via multiagent reinforcement learning; utilize wireless communication technology

(Continues)

TABLE 6 | (Continued)

References	Aim	Gap/problem	Methodology	Findings	Future work
[29]	Introduce a fast algorithm for real-time routing of drones in smart cities using team orienteering problem (TOP)	Current coordination methods are slow and unable to handle real-time changes in smart city scenarios	Combines heuristic routing to maximize rewards with a biased-randomized algorithm to add flexibility and parallel computing for real-time decision-making	High-quality solutions generated in milliseconds, outperforming traditional heuristics	Extend research into dynamic environments with changing inputs and parameters
Drone logistics and delivery					
[21]	To analyze the viability of drones for last-mile delivery in Milan focusing on sustainability, feasibility, and user acceptance	High demand in urban area has caused challenges in smooth supply chains as the adoption of traditional method cause traffic congestion and pollution	Operational design with a "beehive" structure of depots and delivery cycles of 45 min (30 working, 15 charging)	1. Drones preferred for high-value goods but not cost-effective for low-cost items 2. Result shows delivery time reduction for lightweight goods (up to 2.25 kg) 3. Electrically powered, reducing emissions and congestion 4. High initial costs but profitable after years of operation	1. Study end-user recognition and landing site identification 2. Assess environmental impact using multicriteria techniques 3. Investigate hybrid van-drone solutions
[24]	Propose a path optimization method for drone logistics networks focusing on energy efficiency, optimal routing, and reduced delays	Existing networks face energy inefficiency, suboptimal routing, and high delays, particularly in disconnected or sparse areas, reducing delivery effectiveness	Weighted flight path planning (WFPP) optimizes paths by minimizing delays and maximizing delivery ratio, exchanging high-priority packets between drones and considering energy usage and packet priority in routing decisions	WFPP outperforms epidemic and EBR protocols and reduces delivery delays while maintaining low overhead	Future research could focus on examining the cooperative planning among multiple drones

(Continues)

TABLE 6 | (Continued)

References	Aim	Gap/problem	Methodology	Findings	Future work
[25]	Improve large-scale drone traffic in constrained urban airspaces with traffic alignment and segmentation principles	Most research focuses on unconstrained airspaces, leaving a gap in structuring airspace for dense urban areas	Used BlueSky simulator to develop two-way system mimicking streets with altitude-based separation and one-way system with vertical segmentation for directional flow	1. Vertically segmented layers improve safety. 2. One-way system outperforms two-way in safety and stability	1. Investigate dynamic airspace designs for weather adaptation. 2. Include take-off and landing phases for detailed conflict analysis.
Intelligent traffic management and monitoring					
[18]	Address automated pavement distress detection through UAV-generated 3D models	A pavement management system (PMS) is usually high cost and time consuming in data acquisition	UAV with 1 mm ground sampling distance (GSD), and DJI Mavic 2 Pro used. Burst photo with 2-s intervals was taken and transferred to SfM software for distance computation using C2M	The flexible and resource-efficient process successfully calculated the distress metrics	Analyze wider networks and refine distress category metrics
[20]	To enhance urban mobility with the use of UAVs and VLC technologies with a dynamic traffic management system that responds to real-time traffic condition	Traditional ground-based traffic systems fail to handle increasing urban traffic	YOLO was used for congestion detection, VLC technology for establishing communication link between UAVs and traffic light	Accurate congestion detection with YOLO and reliable VLC communication. Reduced dependence on the RF system and reduced energy consumption	Optimize the image processing techniques, enhance system resilience under severe weather, integration of advanced sensors
[23]	Propose a traffic-aware approach to enable drones as mobile edge nodes replacing traditional RSUs	Traditional RSUs face limitations in high-demand scenarios like public events or extreme weather. Fixed architectures with better networks are expensive and infeasible in such contexts	1. Introduced SWIM with: (i) Traffic modeling to estimate conditions (ii) Dynamic deployment to relocate UAVs based on demand 2. Simulations tested SWIM for urban scenarios to analyze coverage and communication overhead	1. UAVs as mobile RSUs provide fast responses in vehicular environments. 2. SWIM achieves full coverage under various conditions with low overhead and reasonable delays. 3. Edges can relocate dynamically when needed	Investigate further on edge replacement and power management

TABLE 7 | Comparative performance metrics from selected studies.

References	Accuracy/error	Efficiency/speed	Scalability/capacity	Resource consumption	Dataset/scenario details
UAV-based object detection					
[16]	• PVswin-YOLOv8s: +4.8% mAP0.5 vs YOLOv8s model • Soft-NMS: 1.1% mAP0.5, + 0.8% mAP0.5:0.95	Detection time • PVswin-YOLOv8s: 6.2 ms • YOLOv8n: 4.4 ms • YOLOv7: 1.9 ms	YOLOv8s chosen for balance of accuracy and efficiency	Model size (PVswin-YOLOv8s): 21.6 MB	VisDrone2019 dataset (640 × 640 px images)
[19]	Detection error: • 11.76% at 2.5–3.0 m • 31.25% at 0.7 m • Accuracy drops above 50 km/h	MobileNet-SSD: real-time, supports 60–90 fps at 480 p Supports both 60 and 90 fps at 480-pixel resolution	Small, fast detector	Low-altitude flights result in reduced battery consumption	Real-world environment at JXUST university intersection; UAV heights 0.7–3.5 m
[22]	High reliability in parking lot counting	Sobel operator: chosen due to low computational cost	Handles rotation and scale invariance	NR	UAV onboard camera images; Otsu's method for variance minimization
[26]	Video processing errors: • RMSE 3.97 px (0.97 m) • Average error 1.77 px (0.43 m) Conflict severity: • PET ≤ 1.0 s (2.7%), PET 1.0–2.5 s (37.9%), PET 2.5–3.5 s (24.5%), PET 3.5–5.0 s (34.9.%)	Object Tracking: 30 fps	NR	NR	Cairo, Egypt: 104,769 frames; 1217 vehicles tracked; 2424 conflicts identified
Flight trajectory and drone traffic control					
[17]	GPS refresh and Kalman filters improved monitoring	CbDMS enables drone management in an real-time environment	Airway-network design improved flow distribution	Optimized for lower total cost (operation, infrastructure, external)	Simulation with SD and Markov models
[27]	Packet loss rate (PLR): • 0.08 Mbps (10 drones) • 0.53 Mbps (100 drones)	Average delay (AD): • 1.59 s (10 drones) • 7.92 s (100 drones) Throughput (THR): • 0.92 Mbps (10 drones) • 0.47 Mbps (100 drones)	Performance validated for 10–100 drones	Energy consumption (ECM): • 42 mJ (10 drones) • 173 mJ (100 drones) Network Lifetime: • 6010 rounds (10 drones) • 4310 rounds (100 drones)	Simulations with 10–100 drones; clustering factors such as energy, CHs, distance

(Continues)

TABLE 7 | (Continued)

References	Accuracy/error	Efficiency/speed	Scalability/capacity	Resource consumption	Dataset/scenario details
[28]	Traffic monitoring: +46.45% more accurate EPOS-balance sensing mismatch: 2.02	2.88% more efficient than Greedy-sensing Computational cost: $O(P \cdot I)$ per agent, $O(P \cdot I \cdot logU)$ for system Communication cost: $O(I)$ per agent, $O(I \cdot \log U)$ for system	Mission inefficiency dropped 83% ⟶ 19% as dispatches increase	Energy: • 271,467 kJ (EPOS-balance) vs 233,484 kJ (min-energy)	Real-world Athens downtown traffic; simulated 1600 × 1600 area, 20k targets, 48 time periods
[29]	Biased-randomized algorithm (BRA) average gap: 0.87% to best-known solution (BKS) Savings-based heuristic average gap: 2.18% to BKS	Heuristics run in 0.13 s on average	Handles large-scale urban routing	NR	34 benchmarks instances from Chao et al. [33] for team orienteering problem (TOP)
Drone logistics and delivery					
[21]	Calibrated alpha coefficients: • −2.185 (van/bicycle) • −0.447 (scooter/drone)	• Each drone perform minimum 18 missions/day • Cycle: 45 min (30 active + 15 charging)	100 drones (960 active + 40 reserve)	Battery capacity and life cycle issues consist major constraint	Case study in Milan, Italy; 1144 daily deliveries
[24]	• Delivery ratio: +25% vs baselines • High-priority packets delivered 20% quicker	Average data delivery delay reduced by 66%	Evaluated 20–100 destinations; 50–250 drones	• Overhead ratio: decreased by 60% vs baselines • Energy inefficiency varies across density	Simulated disjointed network of drones over $10 \times 10\,km^2$ field using “ONE” simulator
[25]	One-way concept showed a lower number of conflicts and intrusions than two-way concept	Drones decelerate by ≥ 50% (to 5 m/s) of their cruise speed (10.3 m/s) to safely execute turns	NR	NR	Urban street network of Manhattan ($59.1\,km^2$) as testing region; 200,000 flights across 60 runs
Intelligent traffic management and monitoring					
[18]	RMSE (control points): • X error 0.0078 m, Y error 0.0082 m, Z error 0.0079 m RMSE (Checkpoints): • X error 0.0095 m, Y error 0.0112 m, Z error 0.0119 m	Survey time: approximately 20 min for the studies section, limited by drone battery life	The workflow can be used to rank road sections within a network based on average distress levels	Drone acquisition cost is marginal compared to survey vehicles and traditional techniques	1000-m road with cracks; 2-s intervals, 70% overlap, RAW format

(Continues)

TABLE 7 | (Continued)

References	Accuracy/error	Efficiency/speed	Scalability/capacity	Resource consumption	Dataset/scenario details
[20]	YOLOv4 achieved 98.08% precision (283 TP, 25 FP)	• Processing: 82.1 fps • Range: 45–150 fps	NR	visible light communication (VLC) chosen for reliability and energy efficiency	Input images resized to 448 × 448 px; traffic in low visibility (fog, pollution) environmental conditions
[23]	Data delivery ratio (DDR): achieved full coverage (almost 1) under medium and high traffic flow rates	Average Dissemination Delay: 200 ms under high traffic demand	Designed or vehicular networks; scalable vs centralized	Broadcast overhead (BO): lower overhead compared to centralized squares	Simulated 4k–9k vehicles/hour; IEEE 802.11p at 5.9 GHz frequency band, 360 m transmission range; 5 Hz data frequency

Note: NR = Not reported in the original study.

geometric reconstruction by utilizing a lightweight MobileNet-SSD model to monitor vehicle speed in specific zones such as schools. This comparison suggests that while photogrammetry offers the precision needed for legal enforcement, lightweight deep learning models are, however, more scalable for general situational awareness in speed-sensitive zones.

Through these studies, it is undeniable that UAV plays a pivotal role in the detection and tracking process, as several studies have pointed out fixed static cameras have limitations in resolving issues such as occlusion, varying illumination conditions, weather, and shadows [19, 22, 40], which may affect the detection accuracy. Traditional fixed-viewpoint object detection methods in detecting vehicle movement also have limitations for vehicle-mounted cameras due to ego-motion, where this results in more processing steps needed to carry out to resolve the issue [42].

Moving from enforcement to safety management, the literature presents two operational phases, namely, preaccident prevention and post-accident response. Shawky et al. [26] addresses the former by utilizing drones to identify “conflict points” and pre-collision behavious, such as tailgating, while analyzing vehicle trajectory data to evaluate intersection safety. In contrast, Alkinani et al. [43] address the postaccident phase by proposing an IoT-based system where UAVs act as first responders to deliver aid. When viewed together, these studies indicate that a holistic smart city system must always integrate Shawky et al.’s predictive analytics to reduce accident frequency with Alkinani et al.’s rapid response architecture to mitigate the accident severity.

4.1.3 | Parking Slot Occupancy Detection

Integrating smart parking slot detection systems in cities not only beneficially resolves the urban traffic congestion issue but also makes full use of available parking spaces in certain places, especially overpopulated areas. D’Aloia et al. [22] advocated for a marker-based approach where the physical markers are printed on parking slots to aid detection. The drone was used for capturing high-resolution footage of parking areas, which were then processed using morphological processing and object warping, along with edge detection and shape contours. This method achieved a cross-correlation coefficient of over 0.992 where this offers exceptional robustness against visual noise. However, this high precision comes at the cost of physical infrastructure modifications.

Contrarily, Guilleum and Sudhanshu [44] developed a novel computer vision approach using CNNs to classify slots as empty or occupied without physical aids. While the solutions provided by D’Aloia et al.’s method are less computationally demanding and more accurate, Guilleum et al.’s deep learning method wins as it offers higher scalability to be deployed over any existing parking lots without requiring manual painting of markers. This suggests future research should focus on bridging this gap to further improve the accuracy of markerless CNN models to match the reality of marker-based systems.

Jiang et al. [45] proposed a parking slot detection system using Mask R-CNN algorithm and applying to the Around-View Monitoring (AVM) system to enhance detection accuracy under various scenarios, while Li et al. [46] aimed to detect the parking slot using classification and line regression with deep

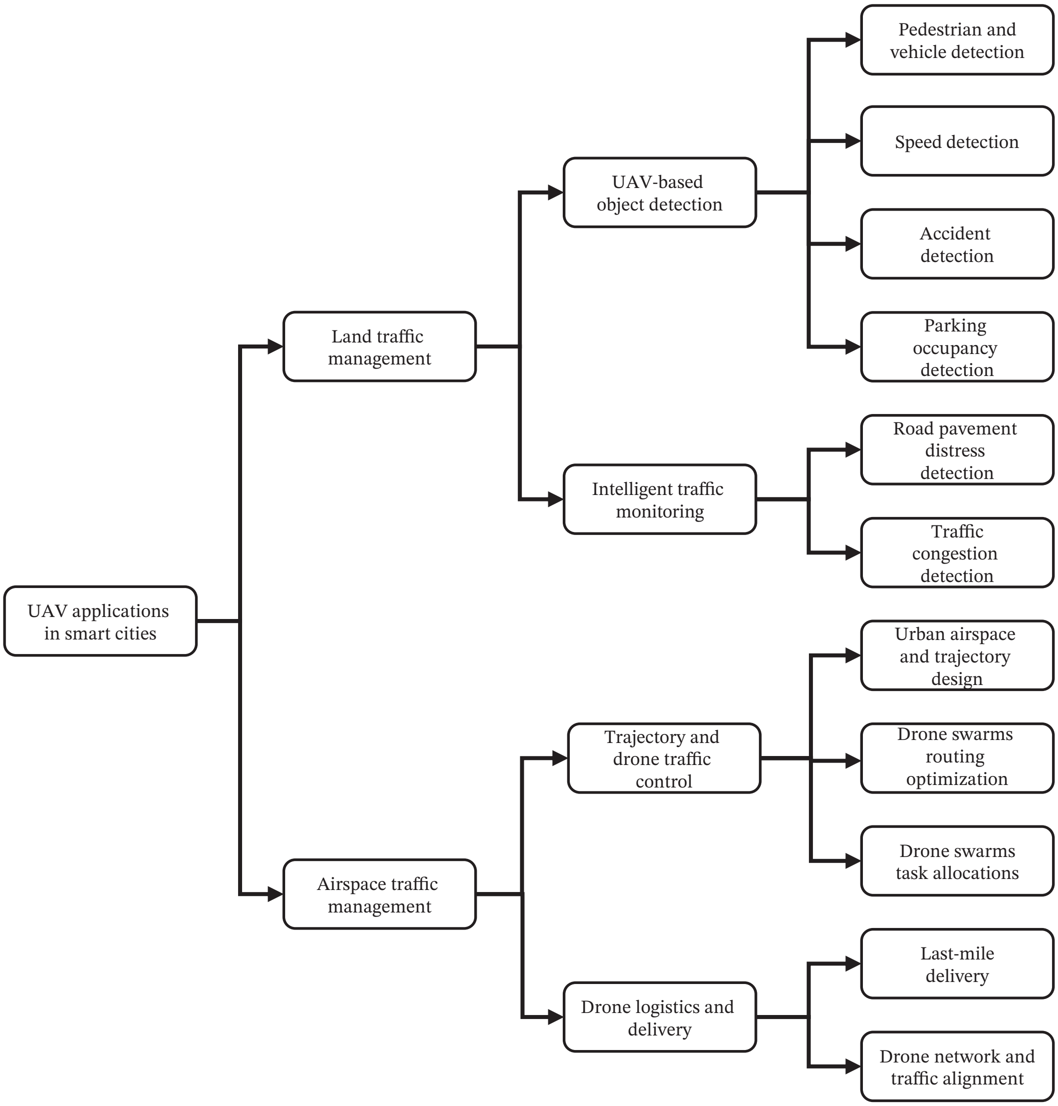

FIGURE 5 | Detailed categorization of UAV applications in smart city traffic management.

CNN (DCNN). Nevertheless, these studies did not employ UAVs in conducting the research. From the articles, it is noticed that the CNN tends to be one of the favored detection algorithms for recognizing empty parking spaces. However, there are still relatively few articles focusing on UAV-based parking slot occupancy detection. Hence, this serves as an underexplored area, which may require more research in the respective field.

4.2 | Flight Trajectory and Drone Traffic Control

An effective drone operation management in urban cities involves certain levels of consideration and breakthroughs such as the need for improving the current air traffic systems to adapt to the increasing complexity of drone operations. Drones are set to become an integral part of smart cities for performing numerous tasks including surveillance [47, 48], delivery [49, 50], emergency response [51–54], and many more. With that being said, there are several initiatives and approaches proposed to further enhance the trajectory planning and overall airspace traffic control system to ensure safety and efficiency. This subchapter explores the potential of fully utilizing drones for drone traffic management, with a focus on designing improved urban airspace systems, coordinating drone swarm routing with task allocation, and optimizing drone routing mechanisms.

4.2.1 | Urban Airspace and Trajectory Design

As drone operations in smart cities scale, the challenge of airspace management has led to two design philosophies, namely, structured fixed corridors versus flexible dynamic grids. Nguyen et al. [17] and Al Ali [55] both aim to organize urban airspace to prevent collisions, but they prioritize different operational constraints. Nguyen et al. advocate for a static, infrastructure-heavy approach by designing specific 3D airways and corridors based on 3D Geographical Information System (GIS) and Global Positioning System (GPS) mapping to minimize the environmental impact. This is because traditional airspace traffic management systems are usually inadequate for vast amounts of drone operations in urban airspace due to high costs and limitations in managing low-altitude drone operations in urban areas. This method creates a “virtual highway” to simplify the

TABLE 8 | Summary of examined studies in 3D road pavement distress detection.

References	Study region	Drone model	Altitude (m)	SFM software
[18]	1-km-long roadway with flat terrain with trees and greenery	DJI Mavic 2 Pro	10	
[37]	100-m-long road segment in University of Palermo, Italy	UAV with GoPro Hero 3 integrated	7	Agisoft Metashape (Previously known as Agisoft PhotoScan)
[38]	Road segment in UiTM and Seksyen 7, Malaysia	DJI Phantom 3 Pro	10 and 40	
[36]	Eight road segments in Fuling District, China	DJI Phantom 4 Pro	—	Pix4Dmapper

trajectory control by limiting drones to predefined paths. Apart from that, a hexagonal grid–based framework built using 4D trajectory representation was developed by Al Ali [55]. The study focused on further improving the conflict detection and planning capabilities for UAVs. This method sought to reduce computational complexity and enhance trajectory following accuracy. Nguyen et al.'s solution provides greater predictability for routine transport, but it lacks adaptability, while Ali et al.'s grid-based indexing method effectively addresses this limitation by prioritizing real-time conflict detection and trajectory planning, making it better suited for high-density dynamic traffic where drones must frequently deviate from set paths to avoid obstacles. This highlights that future urban airspaces must likely to function as a hybrid system utilizing corridors for main transit arteries while employing dynamic grids for last-mile flexibility in complex urban canyons.

4.2.2 | Drone Swarms Optimization and Task Allocation

Overcoming the limitations of traditional existing routing and clustering techniques are crucial in Internet of Drones (IoD) networks. Innovative solutions are needed to solve and manage drone operations in smart cities, especially in terms of data transmission and route selection. In the coordination of drone swarms, the literature reveals a trade-off between computational speed for real-time responsiveness and energy efficiency for mission endurance. Juan et al. [29] prioritize the former, which addresses the need for rapid decision-making in unpredictable smart city environments, by combining biased-randomized algorithms with parallel computing to solve the team orienteering problem (TOP) in routing UAVs in smart cities. This algorithm focuses on overcoming the issue where typical routing algorithms often face: the uncertainty of conditions in urban settings. This would result in longer computational times, hence reducing overall efficiency. As a result, they achieved routing solutions in just milliseconds. This method is ideal for time-critical application such as emergency response where speed is the dominant metric.

However, Ragab et al. [27] and Qin and Pournaras [28] argue that for routine operations, energy efficiency is the primary constraint due to limited battery life. Ragab et al. address this by proposing tuna swarm algorithm–based clustering (TSA-C) and biogeography-based optimization (BBO), which clusters drones to optimize data transmission routes and extend the overall network lifetime. Energy-aware and proportional optimal sensing (EPOS), a decentralized coordination model, was established to coordinate drones in a tree structure to increase the sensing quality while minimizing the energy consumption. These studies indicate that centralized, high-speed algorithms from Juan et al. are necessary for urgent and dynamic interventions, whereas decentralized, energy-aware strategies from Ragab et al. and Qin et al. are robust in long-duration monitoring tasks where longevity outweighs speed.

It was observed that the advancements in airspace traffic management provide huge potential for drones to become a critical part of the urban infrastructure nowadays. The integration of these technologies has shown a major step forward in overcoming common challenges including energy efficiency and power consumption of drones. These innovations ensure that drones can operate more efficiently and in a wider context, more safely in complex urban airspace.

In spite of the considerable advancements in drone flight trajectory optimization and swarm intelligence for urban airspace management, several crucial aspects such as cybersecurity, communication protocols, and privacy still remain significantly underexplored in this context. Recent research mainly focuses on detection algorithms and routing optimizations, while the security of drone operations brings substantial challenges that could undermine the integration of drones into smart city infrastructure [56, 57]. Cybersecurity threats encompass GPS spoofing vulnerabilities, denial-of-service (DoS) attacks, malware injection, and hijacking of drones via insecure communication links. These risks are also exacerbated by having weak encryption and lack of efficient intrusion detection system within drone communication network [57]. Apart from that, the heterogeneity of communication protocols used by drones from Wi-Fi to specialize lightweight protocols such as MAVLink [58] introduces interoperability and security weaknesses that are yet to be addressed systematically. Thus, to realize the full potential of drone integration in smart city traffic management, it is important to introduce a more balanced research focus, where future investigations should prioritize developing secure communication protocols and comprehensive cyber-threat mitigation strategies that complement the existing advances in detection and trajectory control.

4.3 | Drone Logistics and Delivery

In this day and age, the growing trends of urbanization and city development have led to more immense strain on existing urban delivery systems and mechanisms. This may deteriorate current trending issues such as traffic congestion and delays in time, as well as air pollution. In this context, drones stood up to be an excellent option for various sectors to tackle these challenges by providing a fast and sustainable alternative for transporting goods.

By having drone delivery and logistics integrated in smart cities, it serves as a step forward to enhance delivery routes and overall urban mobility. This in turn alleviates ground traffic congestion in the long term [21]. UAVs are becoming an important aspect as well in revolutionizing the logistics system as they can reduce unnecessary costs and increase speed [59]. Hence, this subchapter delves into the advantages and roles drones have provided for further evaluating the convenience of drones in giving a positive impact to urban traffic management.

4.3.1 | Viability of Last-Mile Delivery

Last-mile delivery is commonly referred to as the final stage of the delivery process, where products are transported from a nearby distribution hub to the customer's doorstep [60]. This stage is often the most complex in the supply chain due to challenges such as variable traffic conditions, road disruptions, and the high frequency of short-distance deliveries. It contrasts with the more streamlined and controlled systems used for long-distance transportation.

The integration of drones into last-mile logistics is contingent upon two distinct but interconnected factors: operational efficiency and social acceptance. Borghetti et al. [21] conducted a detailed analysis in Milan city of Italy providing empirical evidence on consumer acceptance and the financial implications of implementing drone delivery services in urban areas. The findings showed that drone delivery has gained immense attention and interest among citizens due to its lower cost and shorter delivery times.

Patro et al. [59] focus on supply chain perspective by utilizing thematic analysis to demonstrate that drones can significantly enhance logistical efficiency by reducing delivery times and operational costs compared to traditional road transport.

With the emergence of drones in last-mile delivery, Jana et al. [61] focus on software optimization by developing an efficient online drone scheduling algorithm for last-mile delivery. The study aims to minimize the number of drones utilized while adhering to the battery constraints. Unlike vast prior research that mostly focused on offline models where customer requests are pre-known, this study emphasized tackling the challenges faced by online requests to manage battery limitations. Tausif [62] conducted an analysis examining the practicality of hybrid delivery systems using mixed-integer linear programming, which improves the performance of the hybrid system. This approach effectively bypasses the existing limitations of drones by physically extending the operational range, suggesting that hybrid ground–air systems offer a more scalable solution for sprawling cities than purely aerial networks.

4.3.2 | Drone Networks and Traffic Alignment in Urban Airspace

Furthermore, a significant conflict exists regarding the structure of the airspace itself. Iranmanesh et al. [24] focus on network flexibility by proposing a weighted flight path planning (WFPP) algorithm to dynamically select optimal routes to maximize the data delivery ratio and energy efficiency. However, Doole et al. [25] aim to investigate the traffic alignment and segmentation principles to highly constrained urban airspace to accommodate higher volumes of drone traffic, prioritizing restrictive safety over flexibility. From these studies, conclusion can be drawn that network researchers always seek flexible paths to optimize overall efficiency, while safety researchers seek rigid and structured pathways for preventing collision, indicating that future regulations must find a compromise between these opposing operational philosophies.

4.4 | Intelligent Traffic Management and Monitoring

While the previous section focused on the fundamental capability of identifying objects, this category assesses the macrolevel application and system integration of drones. The studies reviewed go beyond basic object detection, they investigate how drones integrate with broader smart city infrastructure, such as communicating with traffic lights, acting as roadside units (RSUs), or assessing large-scale road pavement conditions. The focus shifts from "how to detect" to "how to utilize detection data" for more holistic traffic flow control.

By implementing a powerful traffic monitoring approach, it contributes to lower vehicle emissions by minimizing vehicle idle time, which may lead to better air quality in urban cities, especially highly populated areas [63]. For instance, Singapore has established a Smart Mobility 2030 Vision [64], which aims to strengthen the smart city transportation development. Its implementation of the intelligent transportation system (ITS) has effectively reduced road accident occurrences that involve cyclists and pedestrians. An effective system could also reduce the infrastructure damage [65], enhance citizens' satisfaction [63], and increase the overall vehicle speed, which improves traffic flow [66]. This subchapter gradually explores the capability of UAVs in traffic monitoring in smart cities with their respective usage.

4.4.1 | Three-Dimensional (3D) Road Pavement Distress Detection

Table 8 illustrates an overview of the examined studies in 3D road pavement distress detection. Traditional pavement management system (PMS) is struggling due to the high cost and time-consuming nature of obtaining accurate road condition data [67, 68]. Considering this, Roberts et al. [18] and Tan and Li [36] established the baseline viability of the structure-from-motion (SfM) approach, to demonstrate that commercial off-the-shelf drones can achieve millimeter-level accuracy (1 mm GSD) in detecting road anomalies. Their findings confirm that UAVs can effectively replace manual visual inspections, offering faster and safer alternative for updating PMS.

Inzerillo et al. [37] and Saad and Tahar [38] advance this discussion by benchmarking drone-based SfM against the established terrestrial methods. Inzerillo et al. compared UAV-SfM directly with terrestrial laser scanning and reveal that while laser scanning offers higher geometric precision, UAV photogrammetry provides more superior cost-to-accuracy ratio and faster data acquisition for large-scale networks. Saad et al. further refined this methodology by investigating the operational variables, specifically the correlation between flight altitude and detection quality. They found that lower altitudes are critical for detecting subtle depth anomalies like ruts, while potholes are best detected from high elevations.

4.4.2 | Real-Time Traffic Congestion Detection and Management

Beyond passive infrastructure monitoring, UAVs are increasingly proposed as active nodes within the ITS to manage real-time traffic flow. To improve urban traffic management, Garau Guzman and Baeza [20] and El-Sayed et al. [23] address fundamental bottlenecks in current traffic networks, namely, communication latency and network coverage. Focusing on the data transmission method, Garau Guzman and Baeza tackle the communication bottleneck by utilizing the combination of UAVs and visible light communication (VLC) technologies in providing a dynamic traffic management system in terms of real-time congestion detection in smart cities. By leveraging VLC to link drones directly with traffic lights, they offer a solution that bypasses crowded radio frequency (RF) bands to enhance energy efficiency and security.

Bringing attention to the network architecture, El-Sayed et al. focus on the coverage bottleneck inherent in fixed infrastructure. They propose utilizing drones as mobile RSUs that can dynamically position themselves to bridge the connectivity gap during peak congestion or emergencies. Taken together, these studies show a complementary future for smart cities where VLC systems play a crucial role in handling local high-speed data transfer at intersections, while mobile RSUs also ensure network continuity across a broader urban grid [23].

4.5 | Limitations and Challenges in Implementing Drone Technologies

The largest barrier to adopting drone technology is the need for an efficient and effective control system that is able to accommodate the infrastructure. The integration of high volumes of drone traffic in urban airspaces is a significant challenge to overcome [25]. This problem is very relevant for the economic viability of drone usage in commercial delivery systems. However, this issue is one that will affect any smart city system designed to use drones on a large scale. A main area of concern that has not been sufficiently addressed at present is the autonomous pathfinding capabilities of drones within urban environments.

Unlike previous areas of operation, urban environments necessitate upgraded route searching abilities for drones as it is not simply a straight line to go from Point A to Point B. Just like modern car navigation systems, future drone systems also need sophisticated navigation to move around smart cities that take into consideration all the obstacles that will be present. Efficient navigation is also key to reducing the costs of last-mile deliveries. The reviewed research into creating and improving drone navigation has come up with three specific solution methods, swarm intelligence [23, 27, 28], heuristic algorithms [24, 29], and designing specially constrained airspaces for drone usage [17, 25]. The first two methods utilize specially designed route-finding algorithms to control groups of drones deployed in their respective roles. The last method instead aims to create an extensive framework of rules and infrastructure for drone usage, akin to road systems for land-based vehicles, that will be used for all drone operations within the confines of the system's geographical limits.

The widespread use of drones within urban centers, in the role of remote sensing devices, is hampered by the badly congested cities. In these packed conditions, the technologies used for detecting, acquiring, and processing data onboard drones ought to be improved as current technologies are not able to perform primary sensing tasks effectively. Machine learning is a commonly chosen partner for drone technology for remote sensing tasks. Drones, equipped with upgraded object-detection capabilities, have been popularly studied for continuous monitoring of vehicle traffic. In view of this, studies have continuously attempted to reduce congestion and prevent accidents [16, 19, 20, 26]. Other parallel issues, such as parking space management [22] and road pavement maintenance [18], have also been explored for the viability of drone technology-based solutions.

At present, the consensus is that drones have matured enough to be used in many real-life scenarios, but only on a small scale. Both consumers and corporations are assured to embrace the use of drones in everyday civic roles, such as in transforming supply chains [69]. However, the critical success factors (CSFs) for the widespread use of drones are not sufficiently developed at the moment and require further improvements for economic viability. Fehling and Saraceni [70] identified several technical and legal CSFs for drones to become commonplace tools in smart cities.

A critical barrier to real-time traffic monitoring is the size, weight, and power (SWaP) limitation inherent to UAVs. There always exists a fundamental conflict between algorithmic performance and operational endurance. Deep learning models like YOLOv8 used by Tahir et al. [16] require significant GPU processing power. This trade-off is further supported by Bakirci [71] who demonstrated that while YOLOv8 variants offer robust detection accuracy in ITS, their computational demand often exceeds the real-time processing capabilities of standard onboard hardware, necessitating careful model selection based on specific deployment constraints.

Running these heavy architecture models onboard a drone that can deplete the battery rapidly, reducing the flight times from 30 min to less than 15 min, which renders the continuous traffic monitoring impractical. Rather than deploying standard deep learning models, researchers should investigate model quantization and pruning techniques for embedded drone hardware. Developing adaptive inference frameworks that dynamically switch between lightweight detection and heavy processing based on battery levels represents an open opportunity.

Conversely, offloading data to a ground station for processing may save onboard energy but introduce communication latency, which is unacceptable for time-critical applications, such as accident detection [26, 43]. This constraint forces a hard compromise, where researchers must either develop lightweight quantized models like MobileNet-SSD [19] that sacrifice some accuracy for longer battery life, or invest in edge cloud architectures where only critical features are transmitted. Current literature barely addresses this optimization, treating detection accuracy and battery life as separate problems rather than coupled variables.

Since battery technology remains a limiting factor, autonomous inductive charging serves as a key area for future development, where drones can recharge via existing traffic lights or streetlamps during idle monitoring phases. Additionally, for static intersection monitoring, tethered UAV systems should be investigated as a viable alternative to provide unlimited endurance without the weight penalties of heavy batteries.

A significant limitation in the current body of research is the geographic bias toward cities with high infrastructure maturity. The detection models reviewed largely rely on structured road environments characterized by clear lane marking, disciplined traffic flow, and standardized signage. It remains unclear how these findings generalize to cities with mixed traffic cultures or developing infrastructure where vehicle types are highly heterogeneous (e.g., tuk-tuks and motorcycles) and road markings may be faded or nonexistent. Consequently, the high accuracy rates reported in studies [20, 22, 46] may degrade significantly when applied to unstructured traffic environments.

Drone safety is another concern, in both physical and digital space. Drones must have sufficient infrastructure and cybersecurity to perform tasks in urban environments without endangering themselves and others. While the latter is already a subject of research, the former is a matter of continuous development. Once a drone system is in place, threats need to be monitored and concurrently be dealt with just like most modern computer systems. These security concerns can range from common computer viruses to unique device vulnerabilities. From a legal standpoint, all drone-related matters should be the responsibility of a dedicated public governing body. This helps in managing drone usage, within the geographical boundaries of the body's responsibility, by updating rules and legislation, controlling drone traffic, and dealing with any problems related to drones that may occur, such as collisions or other accidents. Finally, a registration system for all individual UAVs allows for better oversight and tracking of drone units to prevent misuse. More research into these facets of drone technology will help transform urban spaces into technologically advanced smart cities.

Beyond technical obstacles, drone technology has raised a number of ethical and societal concerns due to its widespread adoption and deeper integration into various sectors of society. Firstly, the operation of drones has contributed to noise pollution in urban and residential areas primarily as a result of the rapid spinning of propellers [72]. Furthermore, the increased deployment of drones in a swarm to carry out more complex and large-scale tasks such as drone delivery and traffic management has exacerbated the noise pollution issue, making it even more annoying than road traffic or aircraft noise mainly attributed to its particular acoustic feature of pure tones and high-frequency broadband noise [72]. Worse still, prolonged exposure to drone noise could lead to serious mental illnesses such as sleep disturbance, depression, anxiety, mood alterations, and other chronic health problems [73].

Moreover, the fear of privacy invasion has become a major concern of citizens regarding the deployment of drones in the cities. This is largely due to the fact that drones are capable of collecting data and images stealthily as drones are often hardly detected or tracked with naked eyes [74]. Consequently, it has fostered a misperception that drones are being used for surveillance and spying on individuals [74]. This lack of transparency has compounded citizens' concern as citizens do not know whether the data collected by drones would be used or shared inappropriately without consent [75].

Despite these known barriers that are expected to diminish the public acceptance of drone technology, our review reveals a critical gap where the technical studies on drone traffic management rarely consider these negative externalities in their proposed solutions. For instance, while most studies accounted for operational safety during flight and landing, battery optimization, and drone swarm coordination, none included constraints to minimize noise emission from drone operations and emphasized privacy-first deployment in their methodologies.

4.6 | Future Directions to Improve Urban Development

Despite the challenges of drones in revolutionizing the traffic management aspect, the technology possesses substantial potential in aiding other aspects of urban development, attributing to its compact size, high flexibility, and the ability to provide real-time information for faster decision-making with relatively lower cost compared to aircraft such as helicopters. By leveraging the powerful capability of drones, it could help address one of the major problems regarding the constant expansion of urban cities.

Approximately 55% of the global population is living in cities area, which this figure is believed to rise to 68% by the year 2050 [76]. The increasing growth of urban population is deemed to pose a challenge to urban cities in the ability to dynamically accommodate this large population. Hence, by leveraging the technology of drones, it could bring improvements to urban growth by facilitating the process of urban planning, development monitoring, and infrastructure maintenance.

The reason is attributed to the potential of UAVs in facilitating the data acquisition process and enhancing the creation of 3D models. By integrating with various technologies such as SFM, LiDAR, and RGB sensors that augmented the 3D city modeling, drones make it a time-efficient and cost-effective approach in urban planning, development monitoring, and infrastructure maintenance [77, 78]. Without the assistance of drone technology in carrying urban planning, maintaining, and construction monitoring–related tasks, conventional methods in handling these duties are considered to impose a high measurement time and cost, leading to inefficiency.

It has been stressed that the conventional methods in the acquisition of data and urban development monitoring are time-consuming and prone to high error occurrence [18, 78]. Manual acquisition of data is not only an expensive, exhaustive, and time-consuming task, but in the meanwhile, a substantial amount of subjectivity is added to the acquisition process. As a result, these factors hindered urban planning and development processes and infrastructure maintenance.

The integration of drones has significantly improved the speed and cost-efficiency of data acquisition for urban planning. One of the key advantages of drones is their ability to rapidly collect vast amounts of visual and photographic data [78–80]. On top of that, Hu and Minner [77] emphasize the potential of drones to perform real-time and intelligent data processing. These capabilities contribute to reduced measurement time and overall acquisition costs. Furthermore, technologies such as SfM enable the generation of high-resolution 3D models from 2D UAV imagery, providing accurate spatial information that supports effective city planning and management.

Drones also show considerable promise in construction monitoring and infrastructure maintenance due to their efficiency in gathering large-scale data [81–84]. Notably, UAVs could potentially replace traditional airplanes or satellites for urban planning in small- to medium-sized neighborhoods due to their cost-effectiveness. As urban populations continue to grow, further research into the application of drone technologies in these sectors is vital for sustainable development and efficient infrastructure management.

Beyond the technical implementation direction of UAVs, the methodologies of future research must balance efficiency with potential nuisances and privacy concerns caused by drone operations. Researchers should consider integrating frameworks such as "Noise-Aware Path Planning" that restrict flight proximity to residential areas to essential operational tasks to minimize the adverse impact of noise pollution to humans. Furthermore, the idea of Privacy-by-Design should be given enough attention by researchers during the development of UAV integration frameworks for smart city traffic management.

Complementing future technical directions, the evolution of an updated and comprehensive regulatory framework from the aviation authorities and the government is needed [73]. It is worth noticing that there is a crucial connection between the public perception of drone deployment and regulatory factors [73]. A thorough aviation policy including efficient airspace management and specific drone certifications and approval measures is important to meet citizens' expectations. Conversely, a lack of such a policy will have a substantial impact on the public's attitudes toward urban drones as it failed to address public concerns [73]. It is essential for the authorities to take the dual nature of drones, which bring various benefits alongside potential harm, into account while establishing suitable laws and regulations [75]. Lastly, consistency between government regulations and state or local legislation is important as collective effort is the only way of guaranteeing a cohesive and socially responsible implementation of drones in cities [75].

4.7 | Regional and Governance Considerations

UAVs have been widely adopted worldwide over the years with significant uptake in the United States, the Europe Union (EU), Asia, and the Middle East in sectors including civil applications such as agriculture, construction monitoring, and logistics; public safety such as traffic management and disaster response; and surveillance. With the increasing use of UAVs, various law and regulation frameworks have been enacted by government agencies such as the Federal Aviation Administration (FAA) of the United States, the European Union Aviation Safety Agency (EASA) of the EU, the Civil Aviation Administration of China (CAAC), and the General Civil Aviation Authority (GCAA) of United Arab Emirates (UAE), to manage the safety integration of UAVs into national airspace and civil society while mitigating risks related to safety, privacy, and security. Although these frameworks have shared many baseline rules, each country has specific regulations that vary, reflecting distinct national philosophies that shaped the deployment of drones.

Laws and regulations such as weight classification, altitude restrictions, mandatory drone registration, and minimum flying distance from buildings, vehicles, and persons are found across all frameworks [85–88]. These provisions are intended to provide a foundational principle for ensuring aviation safety and reducing public risks, resulting in a convergence of baseline regulatory rules. For instance, while most of the frameworks impose a maximum flight altitude of 150 m above ground level, UAE's GCAA allows the operation of drones up to 1372 m (4500 ft) above ground level [86, 88, 89]. Furthermore, common provisions prohibit the flight of drones within specific distance of buildings, vehicles, and persons [85, 86, 88, 89].

While a common regulatory baseline exists, some jurisdictions have enacted more stringent rules in restricting the flight of drones above uninvolved people or vehicles and crowded areas. For instance, the United States' FAA has restricted drone operation above uninvolved persons unless that person is under a covered structure or inside a stationary vehicle [87]. Similarly, the "Open Category" regulation of the EU's EASA has banned flight operations above uninvolved people [90]. These regulations that prioritize public safety and privacy have reflected the distinct philosophies of the United States and the EU from other global regions through precautionary implementation of drones that has shaped the deployment strategies. However, this has also set a significant barrier to the deployment of drones in the traffic management context as monitoring inherently involves extended periods of operations above populated roadways and uninvolved people, regardless of their movement or shelter. While this cautious approach has indeed mitigated the exposure of public risks, it has obstructed the research and development process and large-scale adoption of drones.

The precautionary regulatory approach evident in the frameworks of the United States (FAA) and EU (EASA) appears to have influenced the research validation methods of researchers. It has been identified in our review that most studies relied solely on simulation testing [17, 23–25, 29]. While most of the studies reviewed did not clearly state regulations as a limitation of their research, the lack of real-world flight testing is consistent with the rigid prohibition of drone flight over crowded areas and moving traffic in Western jurisdictions. This could imply that researchers prefer simulations over real-world testing to avoid the logistical and legal complexities of obtaining flight waivers.

On the contrary, due to China's robust commitment in the research and development of UAVs, it has emerged as the world leader of this swiftly evolving field, showing extraordinary advancements and innovation in the unmanned system [91]. This can be attributed to China's proactive national philosophies of "development-first" in managing unmanned systems evident from its all-inclusive regulatory framework enacted by the CAAC. While the established framework emphasizes safety, it has comprehensive regulations that encourage the expansion, safety, and accountable deployment of drones in the country [91]. As a result, this has facilitated drone research and development, manufacturing and sustaining market growth, paving the way for large-scale deployment.

China's "development-first" philosophy is demonstrated by the presence of real-world deployment in the review. In particular, Ref. [19], was the only study that conducted a real-world flight testing involving the deployment of UAVs at low-altitude airspace above the intersection of JXUST University in China to assess the proposed method's performance in vehicle speed detection. Hence, the capability to validate algorithms in real-world conditions has consolidated China's philosophy regarding low-altitude airspace regulations, which has contributed to a faster transition from theory to practice compared to simulation-based approach.

Likewise, the UAE has portrayed an unwavering commitment in redefining future urban transportation, evident from its considerable regulatory flexibility. For instance, the UAE has begun the mapping of air corridors and the adoption of a strict drone regulatory framework for the integration of piloted and autonomous air taxis and cargo drones into urban areas. This bold step taken by the UAE has shown its dedication in protecting the privacy and safety of the public from the use of drones while pioneering safe, advanced, and sustainable transportation solutions [92].

Although the United States and the EU have adopted strict regulations in restricting drone operations above people and vehicles, their frameworks are not totally inflexible. For example, sector-specific exemptions have been granted to the agriculture industry by the FAA of the United States. The exemptions (17,261 and 18,009) have allowed UAVs to conduct aerial spraying despite baseline prohibitions [93]. These exemptions were rationalized by the importance of agriculture to a country's economy, which has illustrated that regulatory frameworks are adaptable when demanded by national priorities, showing that drone laws are able to evolve with exemptions and waivers that reconcile safety with public benefits. Hence, exemptions for traffic management could be envisioned as drones show substantial potential in traffic monitoring, road safety, and congestion reduction.

A crucial finding discovered in the review is a gap between technical proposal and legal feasibility. Despite the stringent drone regulatory frameworks found worldwide, all the technical studies reviewed did not address legal restrictions in their limitations. This indicates that current research in this field is largely focused on algorithmic optimization while neglecting compliance constraints. This has shown a need for researchers to integrate law regulations into the design of algorithms or frameworks for integrating UAVs into the smart city transportation ecosystem as it is important for achieving real-world deployment viability.

In conclusion, despite a similar regulatory baseline shared globally, regional philosophies have led to divergence in drone deployment strategies. The precautionary approach of the United States and the EU has safeguarded public safety and privacy, but it has slowed down the large-scale adoption of drones in traffic management. Conversely, China's high commitment and proactive approach have enabled the rapid innovation and deployment of drone technology. Looking ahead, the development of UAV traffic management systems and low-altitude corridors may open a structured pathway where safety and innovation are reconciled for the integration of drones in smart city traffic management.

4.8 | Case Study

The deployment of drones in traffic management is no longer a research idea or pilot studies; its implementation has been seen in many cities worldwide. With the characteristics of low cost, high mobility, and compact design, drones equipped with high-resolution cameras, sensors, and even infrared and night vision capabilities have contributed to an adaptive traffic control system, faster response of emergency road events, and more efficient monitoring of traffic violations [94, 95].

One remarkable example of the integration of drones in smart city traffic management is Dubai, where the police deploy drones in monitoring heavy delivery truck violations [95]. On the road, heavy delivery trucks are dangerous entities due to their enormous size and weight, which could increase the risk of extremely severe road accidents when the truck is overloaded, operating during prohibited driving hours, fatigue driving, and more. It is a time-consuming and inefficient task for the inspectors to inspect the delivery truck due to its enormous size that has set a barrier in thorough inspection. Hence, the Roads and Transport Authority (RTA) department, which belongs to the Dubai Police Force, has started to deploy drones to inspect heavy delivery trucks for violations [95]. Nine personnel were trained to carry out the task, which would be dangerous and hard if done manually. Throughout one year of implementation period, more than 300 inspections were carried out with a total of 580 flying minutes recorded and 48 offenses reported [95]. The increased operational efficiency of heavy delivery truck inspection using drones has contributed to a reduction in truck violations and has eased the workload and difficulties faced by road inspectors while inspecting the parts of heavy vehicles, demonstrating drones' capability in seamless integration to a city's broader traffic enforcement framework [95]. Similar initiatives have been seen in India. Fourteen drones were deployed by the police across major intersections in the city of Vijayawada to monitor traffic congestions and violations in real time [96]. This implementation has not only helped the police in reducing traffic congestion but has allowed quicker responses to emergency situations without burdening traffic personnel [96]. Similarly, the police in China have also deployed drones for traffic management and accident response in the district of Yangpu, Shanghai [97]. In road accident scenarios, drones are able to be deployed to the scene for data collection from the vehicles involved within mere minutes. This deployment has reduced accident resolution time by over 40% and increased traffic flow efficiency by 35%, showcasing drones' potential for improving smart city traffic management [97].

In short, these cases demonstrate that drones are no longer a futuristic concept but a practical tool that can be embedded in the smart city traffic management ecosystem. Although various challenges remain, such as urban airspace traffic management, data and citizens' privacy, noise pollution, and battery limitations, the successes in Dubai, Vijayawada, and Shanghai have demonstrated the transformative role drones could play in enhancing the safety, efficiency, and adaptability of traffic management.

5 | Conclusion

In conclusion, this review paper has presented a thorough analysis on exploring the integration of drones in the context of smart city traffic management, along with the underlying challenges in implementing those technologies and possible future research directions. In this systematic review, 14 main research publications were carefully selected from databases such as Scopus, ScienceDirect, SpringerOpen, ACM Digital Library, and EBSCO. By analyzing these studies, it was observed that the implementation of UAVs in enhancing smart cities traffic management has been a rising and trending topic, and it is able to give a positive impact to society as well with its ability in managing both ground traffic and airspace traffic. These advancements indicate that drones have exhibited high potential in strengthening and contributing to a more connected and intelligent urban infrastructure.

The review prioritized studies with demonstratable technical validity, resulting in a focused corpus of 14 studies. While this rigorous filtering excludes a large volume of purely conceptual literature, it ensures the findings presented here are grounded in verifiable engineering evidence rather than theoretical speculation. Apart from that, this study provides a general overview of how drones can be applied in smart cities, without focusing on specific countries or geographical contexts, as many of the reviewed publications do not clearly indicate the regions where these technologies are implemented.

Future research could delve into exploring a regional-based implementation of drones in smart cities. This would give a more detailed analysis on how different regions adapt and develop their drone technologies according to their local laws and regulations. Besides that, it is also recommended to examine the ethical considerations and implications of employing drones in smart traffic monitoring as they consist of some potential privacy concerns [98, 99] due to their nature of being able to capture in both public and private areas.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. *(Supporting Information)*

Graphical Abstract: A comprehensive overview of the systematic methodology and thematic structure employed in this review. It is designed to offer transparency regarding the literature selection process and a roadmap of the core analytical pillars discussed in the paper.